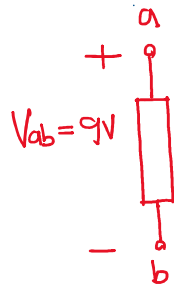
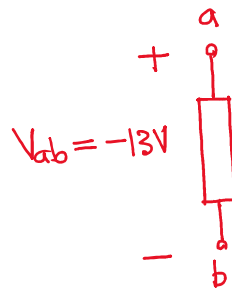
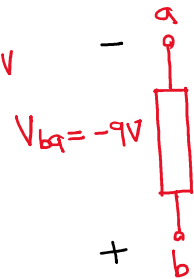


$$V_{ab} = -V_{ba}$$



Find V_{ba} and sketch the voltage

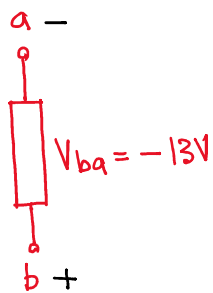
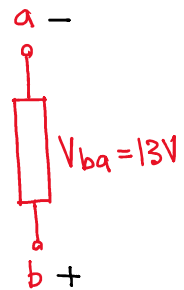
$$V_{ba} = -V_{ab} = -9V$$



Find V_{ba} & sketch

$$V_{ab} = -13V$$

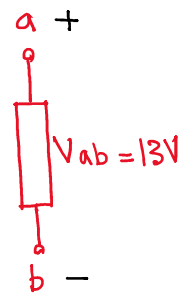
$$V_{ba} = -V_{ab} = 13V$$



Find V_{ab} , & sketch.

$$V_{ba} = -13V$$

$$V_{ab} = -V_{ba} = +13V$$



Steps to determine Electrical Power Supplied/Absorbed by an element.

Check the given current direction:

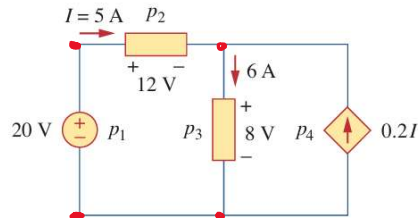
Is it entering the positive terminal of the element?

If Yes,

then multiply the value of the current and the voltage to determine power. If it is positive, power is absorbed and if it is negative, then power supplied.

If No

Assign a current yourself so that it enters the positive terminal. Find the value of the current and multiply it with the voltages to determine power. If it is positive, power is absorbed and if it is negative, then power supplied.

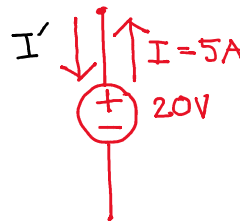


Calculate the power supplied or absorbed by each element in Fig. 1.15.

P_1 :

Assign I' so that it enters the +ve terminal

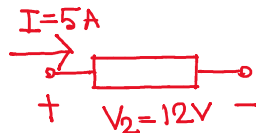
$$I' = -I = -5A$$



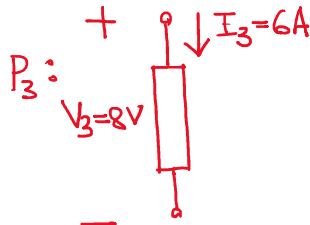
$$P_1 = V_1 I' = 20 \times (-5) = -100W$$

(Supplying)

P_2 :

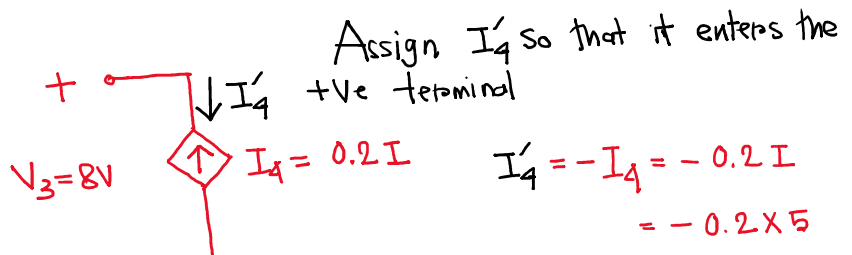


Current I entering +ve terminal
 $P_2 = V_2 I = 12 \times 5 = 60W$
(Absorbing)



Current I_3 entering +ve terminal
 $P_3 = V_3 I_3 = 8 \times 6 = 48W$
(Absorbing)

P_4 :



Assign I_4 so that it enters the +ve terminal

$$I_4 = -I_4 = -0.2I = -0.2 \times 5$$

$$V_3 = 8V$$

$$I_4 = -0.2 \times 5 = -1A$$

$$P_4 = V_3 I_4'$$

$$= 8 \times (-1)$$

$$= -8W \text{ (Power Supplied)}$$

$$P_1 + P_2 + P_3 + P_4 = -100 + 60 + 48 - 8$$

$$= 0$$

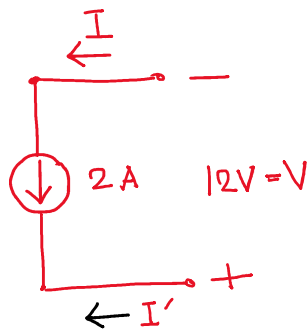
$$\text{Power Supplied} = P_1 + P_4 = -100 - 8 = -108W$$

$$\text{Power absorbed} = P_2 + P_3$$

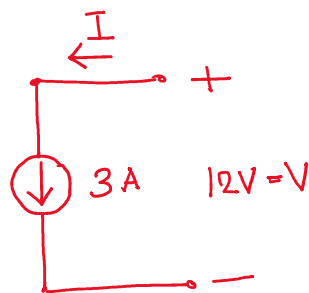
$$= 60 + 48$$

$$\text{Power Supplied} = 108W$$

$$= \text{Power absorbed}$$



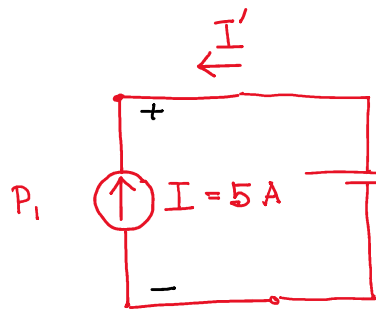
I entering $-ve$ terminal. So Assign a current I' so that it enters the $+ve$ terminal. $I' = -I = -2A$

$$P = VI' = 12 \times (-2) = -24W \text{ (Power Supplied)}$$


$I = 3A$ enters $+ve$ terminal.

$$P = VI = 3 \times 12 = 36W$$

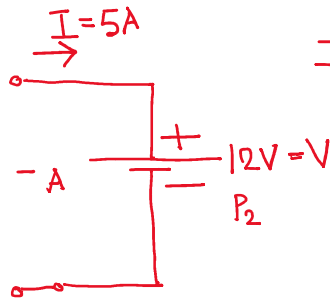
Power P_1 : (Current source)



I entering $-ve$ terminal. So Assign a current I' so that it enters the $+ve$ terminal. $I' = -I = -5A$

$$P = VI' = 12 \times (-5) = -60W \text{ (Power supplied)}$$

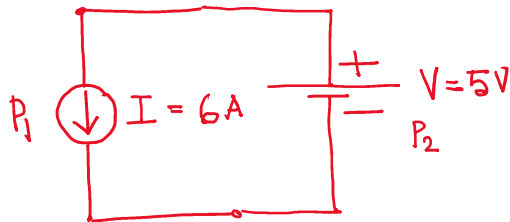
Power P_2 : (12 Volt source)



$I = 5A$ enters $+ve$ terminal.

$$\text{So, } P_2 = VI = 12 \times 5 = 60W \text{ (Absorbing)}$$

Determine, P_1 & P_2 . Which source Absorbing power & which one delivers power?



Determine, P_1 & P_2 . Which source Absorbing power & which one delivers power?

